

Quick Check

Lesson 2: Elements and Compounds of Life: What Is Wanjiru's Lunch Made Of?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 2

Wanjiru's meal has ugali (about 70 g carbohydrate), beans (about 20 g protein and 15 g carbohydrate), sukuma wiki (about 3 g protein plus fibre) and cooking oil (about 10 g fat). What is the total mass of carbohydrate in the meal?

A 70 g

B 85 g

C 105 g

D 95 g

Mr Otieno heats a small piece of dry maize flour in a flame and it turns black. Table salt heated in the same flame does not turn black. What does the black residue show?

- A** The maize flour contains salt, which turns black when heated
- B** The maize flour contains carbon, which is present in organic compounds
- C** The maize flour contains iron, which turns black when heated
- D** The maize flour contains water, which turns black when heated

Starch (a carbohydrate) is made of C, H and O. Which element is found in proteins, such as those in beans and milk, but is NOT found in carbohydrates like starch?

A Oxygen (O)

B Nitrogen (N)

C Hydrogen (H)

D Carbon (C)

Which list contains ONLY organic compounds found in Wanjiru's meal?

- A** Protein, calcium salts, fat, starch
- B** Starch, water, fat, vitamin C
- C** Water, iron salts, calcium salts, protein
- D** Starch, protein, fat, vitamin C

Water is found in every living thing and in all foods, yet it is placed in the inorganic column. Why?

- A** Water is a liquid and organic compounds are always solids
- B** Water is not a chemical, so it cannot be organic
- C** Water is made of only hydrogen and oxygen and has no carbon bonded to hydrogen
- D** Water is not found inside cells

Wanjiru's blood glucose rose after she ate a meal containing about 70 g carbohydrate in ugali, about 20 g protein in beans and about 10 g fat from cooking oil. Which claim and evidence is best supported by this data?

- A** Water is most likely the cause, because it is found in all foods
- B** Protein is most likely the cause, because beans contain 20 g of it
- C** Fat is most likely the cause, because cooking oil is added to the meal
- D** Carbohydrate is most likely the cause, because the meal contains about 70 g of it in ugali, more than any other nutrient

In the class model, the first box 'Food chemicals' is filled in, but the box 'Glucose in blood' is left as a question mark. Why?

- A** We know the kinds of chemicals in the lunch, but we do not yet know which one becomes glucose or how
- B** Carbohydrates have no link to the glucose rise
- C** Glucose is never found in blood
- D** The model is finished and no more boxes are needed

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).